

METAARCHETECH

DNA

Design System Guide · v1.0 · 2026-04-18

INTERNAL REFERENCE · NOT FOR DISTRIBUTION

設計系統指南 — 對內參考文件

{ 'SHOWCASE' }	visustwin-showcase · Desktop / Laptop demo
{ 'WEBCONTROLLER' }	WebController · iPad landscape touch
{ 'WELLTEK' }	welltek-twin · 16:9 full-bleed TV dashboard
{ 'QUARTZ' }	metaarchetech-site · Web responsive
{ 'OMNIVERSE' }	Omniverse extension · Kit UI — colour only

Design Philosophy

ZINC BASE · SIGNAL ACCENT · HUD VOCABULARY · FLOATING PILL · MOTION-RESTRAINED

INTENT

Build a single visual language across all platforms — desktop demo, iPad controller, TV dashboard, web notebook, Omniverse extension. Every port should feel like the same product, not a loose family of look-alikes.

在所有平台上建立統一的視覺語言——每個移植版本都應該感覺像同一產品。

PROBLEM

Fragmented design across five codebases leads to inconsistent brand perception, duplicate token definitions, and friction onboarding new contributors. This guide is the single source of truth.

跨五個代碼庫的設計碎片化導致品牌感知不一致、token 重複定義，以及新貢獻者入門困難。

SCOPE

DNA covers colour, typography, motion, and component primitives. It does NOT prescribe layout grids, business logic, or platform-specific interaction patterns (those live in each repo's own spec).

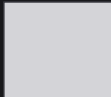
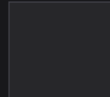
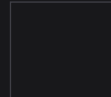
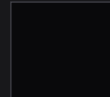
DNA 涵蓋色彩、字體、動效和原始組件。不規定佈局網格或平台特定交互模式。

§02 COLOUR SYSTEM

Colour System

The palette is zinc-based (true steel grey, no warmth) with a single signal-green accent. Dark mode uses a true mirror-flip across zinc-500. Signal greens never change — they are the brand promise.

Zinc Neutrals — Full Scale

										
#FAFAFA zinc-50	#F4F4F5 zinc-100	#E4E4E7 zinc-200	#D4D4D8 zinc-300	#A1A1AA zinc-400	#71717A zinc-500	#52525B zinc-600	#3F3F46 zinc-700	#27272A zinc-800	#18181B zinc-900	#09090B zinc-950

Usage Rules

USE zinc-900 / zinc-950

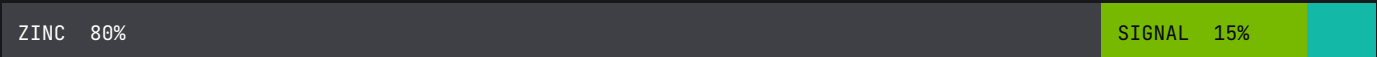
USE zinc-700 / zinc-800

USE zinc-400 / zinc-500

USE zinc-50 / zinc-100

AVOID mixing warm greys

- Page backgrounds, deep card layers
- Component surfaces, modal backgrounds
- HUD labels, metadata, dim descriptors
- Light mode bg only — never on dark pages
- Any grey with a warm tint breaks the steel aesthetic



§02 COLOUR SYSTEM (CONTINUED)

Signal Green — Brand Accent



#9DD11F	#76B900	#5C9300
signal-400	signal-500	signal-600
Dark mode label override	PRIMARY ACCENT — never change	UI kicker labels, section heads

BRAND LOCK — signal-500 #76B900 is THE accent. Never replace with another green.

Semantic Colours



#14B8A6	#F59E0B	#F43F5E	#8B5CF6
teal-500	amber-500	rose-500	violet-500
Environmental data, ambient context	Attention, warnings, caution states	Danger, errors, destructive actions	Reserved / special states, not general use

Colour Do / Don't

- DO Use signal-500 as the single accent on any page or surface.
- DO Use zinc-800 / zinc-700 for layered card surfaces.
- DO Keep semantic colours for their named meaning only.
- DON'T Use multiple accent colours simultaneously.
- DON'T Use warm greys or beige tones anywhere in the UI.
- DON'T Apply signal green to error or warning states.

Typography System

Space Grotesk

LATIN / NUMBERS / HEADINGS

Geometric retro-future grotesque. Weights 400, 500, 600, 700 (700 cap on Google Fonts). Tight letter-spacing on display (-0.04em). Used for all Latin titles, data readouts, UI labels.

wt 400 500 700 · ls -0.04em display · ls 0 body

設計系統指南

CJK / TRADITIONAL CHINESE

?????? variable font. Weights 200-900. Modern HK grotesque with disciplined terminals. Fallback: Noto Sans TC → PingFang TC → Microsoft JhengHei. Slightly reduced em-size vs Latin (CJK glyphs fill box 6-8% larger).

???????? 1.55x · ?? 0.05em

JetBrains Mono

MONOSPACE / HUD LABELS / CODE

All HUD vocabulary. Use ONLY for: structural metadata, sensor readouts, sequence counters, code blocks. Never for body prose. All-caps + tracked (0.06em–0.08em) when used as HUD.

wt 400 500 · UPPERCASE · ls 0.06em–0.08em · tnum lnum

Type Scale

TOKEN	SIZE	LH	LS	USAGE
display	36–48pt	0.95	-0.04em	Page hero, cover title
title	24–32pt	1.05	-0.02em	Section title
heading	16–22pt	1.15	-0.01em	Sub-section heading
body	9–11pt	1.5	0	Prose, descriptions
caption	8pt	1.4	0	Image captions, footnotes
label	8–9pt	1.3	0	UI labels, input fields
hud	7.5–11pt	1.0	0.06em	JBMono · ALL CAPS · metadata
hud-sm	7–8pt	1.0	0.08em	JBMono · tight HUD sub-labels

§04 HUD VOCABULARY

HUD Vocabulary

HUD (Heads-Up Display) text is the structural skeleton of the DNA. It communicates metadata, sequence, and system state — never prose. Rule: if it can be explained in a sentence, it is body. If it is a label, counter, or state flag, it is HUD.

HUD 文字是 DNA 的結構骨架，傳達元資訊、序列和系統狀態，從不用於段落文章。

FONT-FAMILY	JetBrains Mono
TEXT-TRANSFORM	UPPERCASE
LETTER-SPACING	.hud 0.06em · .hud-sm 0.08em
FONT-SIZE	.hud 11px · .hud-sm 10px (absolute, not em)
FONT-FEATURE	"tnum", "lnum" – tabular + lining numerals

HUD Pattern Examples

LIVE STATUS	{ LIVE SENSOR READOUT · 00:42 }
SECTION KICKER	{ YOU ARE ENTERING } EXPERIENCE SEQUENCE
WEEK COUNTER	WEEK 042 / 120 · SPRINT 03
MODULE BRACKET	{ VISTWIN } /02
SPEC TABLE HEADER	SPECS ——— { MODULE-ID }
PAGE HUD	METAARCHETECH DNA · §04 · PAGE 07 / 12
CJK MODULE	{ 资源 } · 04 RESOURCES

HUD vs. Body — Decision Rule

USE HUD FOR	USE BODY FOR
• Sequence counters • Status flags • Module IDs • Timestamps • Section kickers	• Descriptions • Explanations • Prose content • Labels > 5 words • User messages

Primitives

Primitives are the atomic components shared across all DNA ports. Always use DNA primitives before creating custom components. Each primitive has a defined DNA signature — deviation breaks the system.

Pill Header (Glass Nav)

- DNA SIG** floating · top-center · rounded-2xl · backdrop-blur · border-zinc-700/30
- USE** Primary navigation in all web / showcase ports
- COMBINE** Logo · nav links · theme toggle · HUD kicker on far right
- DON'T** Do not use on TV dashboard (no hover). Replace with static top bar.

Glass Card

- DNA SIG** zinc bg layer · border zinc-700/30 · backdrop-blur · slight inner-shadow
- USE** Content containers, capability cards, feature tiles
- COMBINE** Section Label (header) + body prose + HUD spec list (footer)
- DON'T** Never stack more than 3 glass layers — readability degrades.

HUD Ticker

- DNA SIG** JBMono · ALL CAPS · zinc-400/500 · 0.06-0.08em tracked
- USE** Live data readouts, status indicators, sequence counters
- COMBINE** Live dot (signal-500 pulse) + HUD text + optional timestamp
- DON'T** Never use for user-readable prose. Keep under 40 chars per ticker.

Section Label

- DNA SIG** border-t signal-500 · kicker text S6-Bold · HUD right-align counter
- USE** Opens every major content section — sets reading context
- COMBINE** Live dot + KICKER (left) + { MODULE } / O.N (right)
- DON'T** Do not repeat within the same visual block.

§05 PRIMITIVES (CONTINUED)

Sequence Card

- DNA SIG
- hierarchy number (display) + label (SG-Bold) + sub (HUD)
- USE
- Ordered capabilities, roadmap items, numbered features
- COMBINE
- Display number /ON + heading + spec table below
- DON'T
- Do not use without the hierarchy number — it is the identity.

Option Tile

- DNA SIG
- frosted zinc surface · hover: scale 1.025 · signal border on select
- USE
- Selection grids in WebController and showcase demo
- COMBINE
- Icon + HUD label + optional sub-description
- DON'T
- Avoid on TV dashboard — no hover states. Use highlighted list instead.

Scrubber Strip

- DNA SIG
- year HUD label + prev/next buttons + track bar + signal thumb
- USE
- Timeline navigation in showcase and welltek ports
- COMBINE
- Year counter (HUD) + arrow buttons (zinc-700 bg) + progress track
- DON'T
- Not for free-range time selection — DNA scrubber is step-based only.

Shader Background (Soft Flow)

- DNA SIG
- WebGL plane · low-freq noise · accent-reactive to signal-500
- USE
- Full-page atmosphere layer in showcase / hero sections
- COMBINE
- z-index -1 behind all content · opacity 0.6-0.8 · no text on shader
- DON'T
- Never on TV dashboard (GPU budget). Use flat zinc-900 with subtle CSS gradient.

Live Dot

- DNA SIG
- 6px circle · signal-500 fill · 2.2s pulse animation · box-shadow glow
- USE
- Live status indicators, section kickers, ticker labels
- COMBINE
- Always precedes HUD text in section kickers
- DON'T
- Do not use for non-live decorative purposes. Pulse = "this is live data."

Row Accent

- DNA SIG
- 2px left border · scaleY(0→1) on hover · signal-500 · 0.65s expo-out
- USE
- Table rows, list items in capabilities and spec tables
- COMBINE
- Pair with row-tint (signal @ 2.5% opacity on hover bg)
- DON'T
- Skip on TV dashboard (no hover). Remove with prefers-reduced-motion.

§06 MOTION

Motion

Motion is structural, not decorative. Every animation must serve a purpose: directing attention, confirming state change, or communicating hierarchy. A single easing function (expo-out) is used throughout — consistency over variety.

動效是結構性的，不是裝飾性的。每個動畫都必須服務於目的。

```
MASTER EASING — EXPO-OUT
cubic-bezier(0.16, 1, 0.3, 1)
```

Animation Catalogue

ENTER STAGGER	0.6-0.7s	opacity 0→1 + y 12→0, 0.04s delay per item (framer-motion)
HOVER SCALE	—	whileHover: scale 1.025 · whileTap: scale 0.95
LAYOUT MORPH	0.4s	framer-motion layoutId — shared element transitions
COLOUR TRANSITION	500ms	CSS transition for bg/color changes on theme or state
ROW ACCENT DRAW	0.65s	scaleY 0→1 on left border — section row hover
LINK UNDERLINE DRAW	0.55s	background-size 0→100% — left-to-right underline
ARROW TRANSLATE	0.55s	translateX 0→4px on parent: hover
HERO ENTRANCE	0.8-0.9s	Reveal (clip-mask y 110%→0%) or FadeUp (opacity + y 14)
HEADER GLASS BLUR	0.3s	backdrop-blur + bg-opacity on scroll — 300ms ease
LIVE DOT PULSE	2.2s ∞	box-shadow 0→8px→0 ease-out — always signal-500

DON'T — Prohibited Motion Patterns

- ✗ Spinning loaders or decorative rotation — use a progress bar or skeleton instead.
- ✗ Bouncy spring physics (overshoot > 5%) — expo-out is the ceiling.
- ✗ Duration > 1.0s for UI feedback — transitions above 700ms feel sluggish.
- ✗ Multiple simultaneous entrance animations in the same viewport region.
- ✗ Animations that do not respect prefers-reduced-motion.
- ✗ Colour-cycling, shimmer, or gradient-animate on background surfaces.

```
@media (prefers-reduced-motion: reduce)
  All durations collapse to 0.01ms. All framer-motion variants must check useShouldAnimate().
```

§07 DEVICE TIERS

Device Tiers

DNA is deployed across five distinct platform contexts. Each tier inherits the full colour and type system but applies different constraints on layout, motion, and interaction. Always confirm your target tier before implementing.

T1

visustwin-showcase

DESKTOP / LAPTOP DEMO

MacBook 14/16 · 2560×1664 HiDPI / 1440p

- Full primitives set — glass, shader bg, all motion
- Hover states and keyboard nav required
- Framer-motion scroll-trigger, Reveal, stagger — all enabled

WHY: Primary brand demonstration surface — no capability restrictions.

T2

WebController

IPAD LANDSCAPE TOUCH

iPad Pro 12.9" · 1366×1024 (landscape)

- Touch targets >= 44pt — no small hit areas
- Hover states omitted (touch device has no hover)
- Controls must occupy >= 80% of screen real estate

WHY: Operator interface for live demos. Touch-first, control-first — no decorative bloat.

T3

welltek-twin

16:9 TV DASHBOARD — NO SCROLL

1920×1080 or 2560×1440 full-bleed TV

- ZERO scrolling — all content must fit in viewport
- If content overflows: integrate / carousel / tab — never overflow
- No hover states, no backdrop-blur (performance budget)

WHY: Permanent display installation. No operator interaction, no scroll capability wired up.

T4

metaarchetech-site (Quartz)

WEB RESPONSIVE — QUARTZ

320px–1600px responsive

- Quartz SCSS overrides only — do not fight the Quartz component tree
- Colour tokens via CSS custom properties (no Tailwind)
- Motion limited to CSS transitions — no framer-motion in Quartz

WHY: Knowledge vault / notebook. Content readability first, DNA aesthetics second.

T5

Omniverse extensions

KIT UI — COLOUR ONLY

NVIDIA Omniverse Kit extension panels

- ONLY colour tokens are shared — layout, motion, primitives do NOT apply
- Kit UI controls must be used — DNA primitives cannot render in Kit
- Use signal-500 / zinc scale inside Kit's colour parameter system

WHY: Kit has its own rigid layout and widget system. Colour is the only safe cross-port token.

§08 DNA CHECKLIST

DNA Port Checklist

Run through this checklist for every new port, new component, or design review. Each item maps to a specific section in this guide.

每個新的移植版本、新組件或設計審查都應執行此清單。

COLOUR

- ☐ Only palette colours used (zinc + signal + semantic only)
- REF §02
- ☐ No warm greys, no off-brand greens, no arbitrary hex values
- REF §02
- ☐ Signal-500 is the only primary accent — not signal-400 or signal-600
- REF §02
- ☐ Semantic colours (teal/amber/rose/violet) used only for their named purpose
- REF §02
- ☐ Dark mode: zinc-900 bg confirmed, signal greens unchanged
- REF §02

TYPOGRAPHY

- ☐ Headings use Space Grotesk Bold — not system-ui or Arial
- REF §03
- ☐ CJK text uses Noto Sans HK or Chiron Hei HK — not fallback only
- REF §03
- ☐ Type scale respected: display/title/heading/body/caption/label/hud
- REF §03
- ☐ No custom font sizes outside the scale without justification
- REF §03

HUD VOCABULARY

- ☐ HUD text uses JetBrains Mono — not SG or NotoHK
- REF §04
- ☐ HUD is ALL CAPS + tracked (0.06em+) + tabular numerals
- REF §04
- ☐ HUD used only for metadata/state — not for prose
- REF §04
- ☐ Dot separators (·) and bracket notation used correctly
- REF §04

MOTION

- ☐ All animations use cubic-bezier(0.16, 1, 0.3, 1)
- REF §06
- ☐ No animation exceeds 1.0s duration for UI feedback
- REF §06
- ☐ prefers-reduced-motion collapses all durations to 0.01ms
- REF §06
- ☐ No decorative spinning, bouncy, or gradient-cycle animations
- REF §06

DEVICE TIER

- ☐ Target tier confirmed before implementation (T1-T5)
- REF §07
- ☐ TV (T3): zero scroll, no hover, min 12px text — verified
- REF §07
- ☐ iPad (T2): touch targets >= 44pt — verified
- REF §07
- ☐ Omniverse (T5): only colour tokens imported, no primitives
- REF §07

PRIMITIVES

- ☐ DNA primitives used before custom components are created
- REF §05
- ☐ Glass Card: backdrop-blur + zinc-800 bg + zinc-700/30 border
- REF §05
- ☐ Section Label: signal border-t + kicker left + counter right
- REF §05
- ☐ Live Dot: only signal-500, only for live data — not decorative
- REF §05

One system.
Five surfaces.
Zinc base. Signal accent. No compromise.

METAARCHETECH DNA · v1.0 · 2026-04-18
author: metaarchetech · authority: visustwin-website
internal reference – not for public distribution